

Stock Market Portfolio Tracker - Project Documentation

1. Introduction

The Stock Market Portfolio Tracker is a web-based application designed to help users manage their stock investments, track market data, and analyze portfolio performance. It provides features such as portfolio management, watchlist tracking, real-time price updates, and visual dashboards. The application supports multiple user roles including Admin and Investor, with secure authentication and role-based access control.

2. Technology Stack

- Backend: ASP.NET Core Web API (.NET 8), Entity Framework Core, SQL Server
- Frontend: React (TypeScript), Axios, Chart.js/Recharts
- Tools: Visual Studio 2022, VS Code, Postman, GitHub, Copilot

3. User Roles & Authentication Flow

The application supports multiple user roles:

- Admin: Can manage users, stocks, sectors, and exchanges.
- Investor: Can manage their portfolio and watchlist.

Authentication is handled via secure login and registration forms. Tokens are issued for session management and role-based access control.

4. Database Schema

The database schema includes the following tables:

- Users: Stores user credentials and roles.
- UserTokens: Manages session tokens.
- Stocks: Master list of stocks.
- UserHoldings: Tracks user investments.
- Watchlist: Tracks stocks users are interested in.
- StockPrices: Stores price data for stocks.

Indexes are created on key columns for performance optimization.

5. API Endpoints

The application exposes RESTful APIs for:

- Authentication: Login, Register, Token validation
- Portfolio: Add/Edit/Delete/View holdings

- Watchlist: Add/Remove/View watchlist items
- Market Data: Fetch current and historical prices
- Admin: Manage users and stocks

6. Frontend Component Structure

The frontend is built using React with TypeScript. Components are structured as follows:

- Auth Components: Login, Register
- Dashboard Components: Summary, Charts
- Portfolio Components: Holdings Table, Actions
- Watchlist Components: Watchlist Table, Filters
- Admin Components: User Management, Stock Management

Axios is used for API integration and state management is handled using React hooks and context.

7. Page-wise Layouts & Operations

Portfolio Page

Displays user's stock holdings with options to add, edit, delete, and view detailed stock information. Includes summary metrics and charts.

Dashboard Page

Shows overall portfolio performance including total investment, current value, profit/loss, and visual charts.

Watchlist Page

Allows users to track stocks of interest with options to add to portfolio or remove from watchlist.

Login/Register Page

Provides secure login and registration forms with role selection and validation.

Admin Panel

Enables admins to manage users, stocks, sectors, exchanges, and view system logs.

8. Charts & Visualizations

The application includes the following charts:

- Pie Chart: Sector allocation of portfolio
- Line Chart: Portfolio value over time
- Bar Chart: Gains and losses per stock

9. Learning Outcomes

This project demonstrates the integration of frontend and backend technologies, secure authentication, RESTful API design, database schema optimization, and data visualization techniques. It provides practical experience in building scalable and user-friendly web applications.

Mock UI Summary

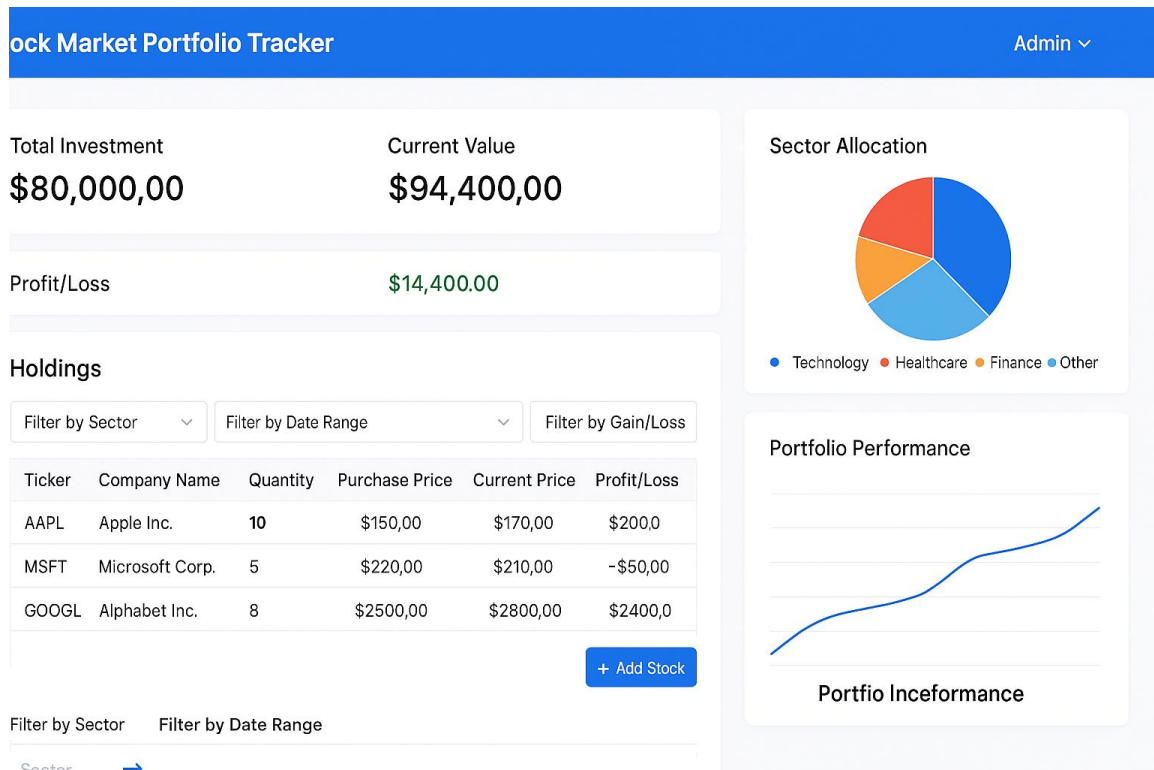
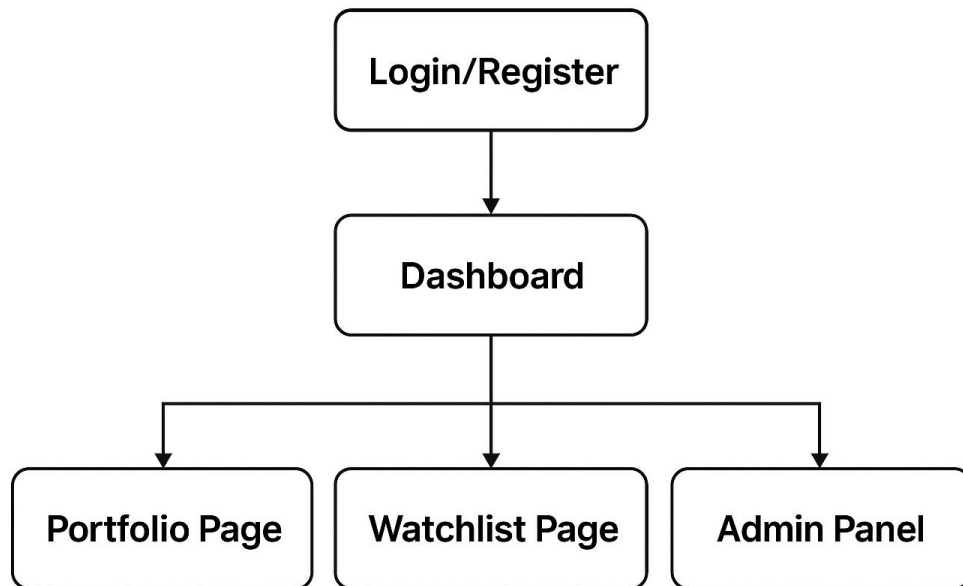


Figure: Combined mock UI layout for all pages.

Website Layout



Login/Register Page Layout

The image shows a mockup of the Login and Register forms. The 'Login' form on the left has a blue header, an 'Email Address' field, a 'Password' field, a blue 'Log in' button, and a link to 'Don't have an account? Register'. The 'Register' form on the right has a blue header, a 'Name' field, an 'Email Address' field, a 'Password' field, a blue 'Register' button, and a link to 'Already have an account? Log in'.

Post-Login Landing Page: Dashboard Layout

Role-Based View

- **Admin Dashboard:** Focused on system-wide metrics and management tools.
 - **User Dashboard:** Focused on personal portfolio insights and market tracking.
-

For Regular Users (Investors)

Header

- Navigation bar with links to:
 - Dashboard
 - Portfolio
 - Watchlist
 - Profile
 - Logout

Summary Cards

- **Total Investment**
- **Current Value**
- **Overall Profit/Loss**
- **Number of Stocks Held**

Charts Section

- **Pie Chart:** Sector allocation
- **Line Chart:** Portfolio value over time
- **Bar Chart:** Gains/Losses per stock

Quick Actions

- Button: "Add New Holding"

- Button: “View Watchlist”
- Button: “Go to Portfolio”

Recent Activity

- List of recent transactions (buy/sell)
- Notifications (e.g., price alerts, stock news)

For Admins

Header

- Navigation bar with links to:
 - Dashboard
 - User Management
 - Stock Management
 - System Logs
 - Logout

Admin Summary Cards

- **Total Users**
- **Total Stocks**
- **Recent Registrations**
- **System Health (optional)**

Management Panels

- **User Management:** Quick links to view/edit users
- **Stock Management:** Add/edit/delete stocks
- **Audit Logs:** View recent system activity

Charts

- User growth over time
- Stock additions/modifications
- System usage trends